

Hieu Nguyen

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EDUCATION

University of Florida | Gainesville, FL

Expected December 2027

Bachelor of Science in Computer Science, Minor in Engineering Innovation

GPA: 3.98/4.0

SKILLS

Languages: JavaScript, TypeScript, Python, Go, C#, C++, SQL, HTML/CSS

Frameworks: Node.js, Next.js, React.js, Pandas, NumPy, .NET

Tools: Linux/Bash, Git, Github, Github Actions, Docker, AWS, Cloudflare, Vercel, Figma, PostgreSQL, MongoDB, Microsoft SQL Server, Postman, Firebase, Jira, Microsoft PowerBI, Claude, Github Copilot, Cursor

EXPERIENCE

Software Engineer Co-Op

January 2026 – August 2026

Collins Aerospace (RTX) - Factory Enablement & Support

Melbourne, FL

- Optimized a Python pipeline for parsing and querying **100,000+** JSON test records, reducing processing time by **80%** and storage usage by **60%** through algorithmic and data-structure improvements.
- Engineered a full-stack troubleshooting application with React, Typescript and FastAPI using historical test data, reducing diagnosis time for failed units by **3+** hours per week across multiple production sites.
- Developed and deployed C# (WPF) desktop applications using the MVVM design pattern and Microsoft SQL Server, enhancing internal software testing workflows and enabling software-version assignments for **500+** production operators.

Technical Director

March 2026 – Present

SwampHacks, University of Florida

Gainesville, FL

- Lead a team of **7** developers building and maintaining SwampHacks' technology platform and website using React, TypeScript, Go, and PostgreSQL, supporting the organizing team and over **1000+** student applicants.
- Architect and maintain the CI/CD and deployment infrastructure with **GitHub Actions**, **Docker**, and **Digital Ocean**, automating releases and ensuring 24/7 uptime for the platform.

Undergraduate Researcher

August 2024 – Present

Prevatt Research Group, University of Florida

Gainesville, FL

- Engineered a machine learning pipeline using PyTorch and Numpy to classify roof types of buildings using **3,000+** aerial images from Google Maps API for hurricane damage assessment studies, achieving up to **80%** prediction accuracy.
- Debugged an open-source Python package for modeling wind-induced damage and loss and visualized simulation results using Python and Matplotlib to validate damage trends across wind speeds and identify code improvements.

Student Mentor

May 2024 – June 2024

Goldberg Gator Engineering Explorers, University of Florida

Milton, FL

- Mentored 10+ high students in engineering and programming fundamentals using Micro:bits and Microsoft Makecode.
- Supported class activities by helping students debug programs, solve technical challenges, and strengthen problem-solving skills.

SOFTWARE PROJECTS

GitFlow | [Devpost](#) | React, FastAPI, Express.js, Docker

September 2025

- 1st place winner** and “Most Creative” project at ShellHacks 2025 for engineering a tool that enables developers to explore the structure of a Github repository with zero set up.
- Developed deployment workflow with Express API endpoints and pipelines aligned with microservices principles; integrated ElevenLabs AI voice summaries for real-time code reviews and insights.

Online Chess Platform | [Github](#) | Rust, React, Typescript, Websockets, Nginx

May 2023

- Built a full-stack chess platform with real-time multiplayer gameplay using React, WebSockets, and a Rust-based chess engine.
- Deployed to a DigitalOcean VPS with Nginx, gaining hands-on experience with DevOps and deployment practices.